

Related Rates

Filling a Tank (Canonical)

Thomas Calculus. Water runs into a conical tank at the rate of $9 \text{ ft}^3/\text{min}$. The tank stands point down and has a height of 10 ft and a base radius of 5 ft . How fast is the water level rising when the water is 6 ft deep?

Howard Anton. Suppose that liquid is to be cleared of sediment by allowing it to drain through a conical filter that is 16 cm high and has a radius of 4 cm at the top. Suppose also that the liquid is forced out of the cone at a constant rate of $2 \text{ cm}^3/\text{min}$.

- Do you think that the depth of the liquid will decrease at a constant rate? Give a verbal argument that justifies your conclusion.
- Find a formula that expresses the rate at which the depth of the liquid is changing in terms of the depth and use that formula to determine whether your conclusion in part (a) is correct.
- At what rate is the depth of the liquid changing at the instant when the liquid in the cone is 8 cm deep?

James Stewart (Exercise). A water tank has the shape of an inverted circular cone with base radius 2 m and height 4 m . If water is being pumped into the tank at a rate of $2 \text{ m}^3/\text{min}$, find the rate at which the water level is rising when the water is 3 m deep.

Oil Spill

Thomas Calculus Problem 46. An explosion at an oil rig located in gulf waters causes an elliptical oil slick to spread on the surface from the rig. The slick is a constant 9 in. thick. After several days, when the major axis of the slick is 2 mi long and the minor axis is $3/4 \text{ mi}$ wide, it is determined that its length is increasing at the rate of 30 ft/hr , and its width is increasing at the rate of 10 ft/hr . At what rate (in cubic feet per hour) is oil flowing from the site of the rig at that time?

Howard Anton. Assume that oil spilled from a ruptured tanker spreads in a circular pattern whose radius increases at a constant rate of 2 ft/s . How fast is the area of the spill increasing when the radius of the spill is 60 ft ?

Smith Minton. An oil tanker has an accident and oil pours out at the rate of 150 gallons per minute. Suppose that the oil spreads onto the water in a circle at a thickness of $\frac{1}{10}''$. Given that 1 ft^3 equals 7.5 gallons, determine the rate at which the radius of the spill is increasing when the radius reaches 500 feet.
